

Hands-on Lab: Optimizing dvdrental

Name: Nguyen Thi Hai Yen

ID: 11247252

Task 1: Influence the Optimizer

```
76  --Hands-on Lab: Optimizing dvdrental
77  -- Force a Sequential Scan by disabling Index Scans
78  SET enable_indexscan = off;
79  EXPLAIN ANALYZE SELECT * FROM rental WHERE rental_date > '2005-05-24';
80  -- Re-enable and compare the Cost and Time
81  SET enable_indexscan = on;
82  EXPLAIN ANALYZE SELECT * FROM rental WHERE rental_date > '2005-05-24';
```

Data Output Messages Notifications

Showing rows: 1

	QUERY PLAN	
	text	
1	Seq Scan on rental (cost=0.00..350.55 rows=16044 width=36) (actual time=0.030..2.578 rows=16044.00 loop...	
2	Filter: (rental_date > '2005-05-24 00:00:00'::timestamp without time zone)	
3	Buffers: shared hit=150	
4	Planning:	
5	Buffers: shared hit=3	
6	Planning Time: 0.216 ms	
7	Execution Time: 3.765 ms	

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Task 2: Join Strategy Analysis.

	QUERY PLAN
	text
1	Hash Join (cost=150.55..545.57 rows=16044 width=122) (actual time=4.977..12.962 rows=16044.00 loops=1)
2	Hash Cond: (r.inventory_id = i.inventory_id)
3	Buffers: shared hit=177 read=7
4	-> Hash Join (cost=22.48..375.33 rows=16044 width=106) (actual time=4.344..9.023 rows=16044.00 loops=1)
5	Hash Cond: (r.customer_id = c.customer_id)
6	Buffers: shared hit=152 read=7
7	-> Seq Scan on rental r (cost=0.00..310.44 rows=16044 width=36) (actual time=0.016..0.692 rows=16044.00 loop...
8	Buffers: shared hit=150
9	-> Hash (cost=14.99..14.99 rows=599 width=70) (actual time=4.316..4.317 rows=599.00 loops=1)
10	Buckets: 1024 Batches: 1 Memory Usage: 72kB
11	Buffers: shared hit=2 read=7
12	-> Seq Scan on customer c (cost=0.00..14.99 rows=599 width=70) (actual time=0.022..4.228 rows=599.00 loop...
13	Buffers: shared hit=2 read=7
14	-> Hash (cost=70.81..70.81 rows=4581 width=16) (actual time=0.598..0.598 rows=4581.00 loops=1)
15	Buckets: 8192 Batches: 1 Memory Usage: 279kB
16	Buffers: shared hit=25
17	-> Seq Scan on inventory i (cost=0.00..70.81 rows=4581 width=16) (actual time=0.018..0.216 rows=4581.00 loops...
18	Buffers: shared hit=25
19	Planning:
20	Buffers: shared hit=399 read=9 dirtied=1
21	Planning Time: 24.776 ms
22	Execution Time: 13.546 ms

Selected Join Algorithm: PostgreSQL chose Hash Join for both steps of the query:

- Step 1 (Inner Join): Hash Join between customer and rental on `r.customer_id = c.customer_id`. The customer table (599 rows) was loaded into memory to build the Hash Table (72kB).
- Step 2 (Outer Join): Hash Join between the result of Step 1 and inventory on `r.inventory_id = i.inventory_id`. The inventory table (4,581 rows) was loaded into memory to build the Hash Table (279kB).

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	QUERY PLAN	
	text	🔒
1	Hash Join (cost=150.55..545.57 rows=16044 width=122) (actual time=1.545..20.412 rows=16044.00 loops=1)	
2	Hash Cond: (r.inventory_id = i.inventory_id)	
3	Buffers: shared hit=184	
4	-> Hash Join (cost=22.48..375.33 rows=16044 width=106) (actual time=0.210..10.291 rows=16044.00 loops=1)	
5	Hash Cond: (r.customer_id = c.customer_id)	
6	Buffers: shared hit=159	
7	-> Seq Scan on rental r (cost=0.00..310.44 rows=16044 width=36) (actual time=0.018..1.609 rows=16044.00 loop...	
8	Buffers: shared hit=150	
9	-> Hash (cost=14.99..14.99 rows=599 width=70) (actual time=0.175..0.177 rows=599.00 loops=1)	
10	Buckets: 1024 Batches: 1 Memory Usage: 72kB	
11	Buffers: shared hit=9	
12	-> Seq Scan on customer c (cost=0.00..14.99 rows=599 width=70) (actual time=0.008..0.060 rows=599.00 loo...	
13	Buffers: shared hit=9	
14	-> Hash (cost=70.81..70.81 rows=4581 width=16) (actual time=1.275..1.289 rows=4581.00 loops=1)	
15	Buckets: 8192 Batches: 1 Memory Usage: 279kB	
16	Buffers: shared hit=25	
17	-> Seq Scan on inventory i (cost=0.00..70.81 rows=4581 width=16) (actual time=0.010..0.359 rows=4581.00 loops...	
18	Buffers: shared hit=25	
19	Planning:	
20	Buffers: shared hit=97	
21	Planning Time: 0.820 ms	
22	Execution Time: 21.584 ms	